

# Conjecture Settling Part I

Me + ChatGPT

*Video Reference:* <https://www.youtube.com/watch?v=PU1LMVGcyXA>

## Introduction

In terms of a Clay problem or any conjecture, I found a plan to use an algorithm that would output, of conjecture  $c$ :

1.  $c$  is a theorem
2.  $\neg c$  is a theorem
3. the ATP ran for  $N$  steps and did not determine either (1) or (2)

Basically, the proof of (1) or (2) would start with the conjunction of all hypotheses and axioms (which may need infinitary logic if that conjunction is countably infinite).

Then every odd step is towards an effort to prove  $c$ . Every even step is towards an effort to prove  $\neg c$ . Perhaps odd steps will only use odd steps before it, but perhaps not. Either way, both branches come from the axioms + hypotheses; so, both even and odd steps are allowed to use all previous steps in the proof, even if that previous step was “intended” to prove the negation of its goal.

This seems to me to be like an ATP having an open mind, and not biased in that it is only trying to prove  $c$ , and then humans take the ATP running for a long time as “evidence” for  $\neg c$ .

But this plan would not have any bias at all. This target-seeking plan is both trying to prove  $c$  and  $\neg c$  while both branches start with the hypotheses + axioms; so, at every step, even and odd, all previous steps can be used, whether or not a human would have foreseen that a lemma in the proof of  $c$  was, at that step, the output of a target-seeking algorithm for  $\neg c$  (or replace  $c$  with  $\neg c$  and  $\neg c$  with  $c$ ).

## Query to ChatGPT

I asked ChatGPT (which knows about my manuscript) what it has to say about this:

## Two Distinct Ideas

I think there are actually two distinct ideas in what you're proposing:

1. A practical ATP search strategy (search for both a proof and a refutation simultaneously).
2. A SIC/Depths-of-a-Simulation interpretation of what it means when neither search succeeds after a large amount of computational effort.

The first idea is standard enough that logicians would recognize it as a form of dovetailing or parallel proof search. The second is where your framework becomes interesting.

## The ATP Perspective

Suppose  $T$  is the theory consisting of all axioms and hypotheses, and  $c$  is a conjecture.

Instead of searching only for a proof of  $c$ , run two searches simultaneously:

- **Search A:** enumerate possible proofs from  $T$  whose conclusion is  $c$ .
- **Search B:** enumerate possible proofs from  $T$  whose conclusion is  $\neg c$ .

At stage  $n$ :

- odd-numbered stages advance Search A,
- even-numbered stages advance Search B.

The machine outputs:

- “Theorem” if it finds a proof of  $c$ ,
- “Refuted” if it finds a proof of  $\neg c$ ,
- otherwise “undetermined after  $N$  steps.”

For a sound and complete proof system:

- If  $T \vdash c$ , eventually Search A halts.
- If  $T \vdash \neg c$ , eventually Search B halts.
- If neither is provable, neither branch halts.

This is essentially a semidecision procedure for both sides simultaneously.

## The Subtle Point You Noticed

The interesting feature is that the search is not psychologically “biased.”

Humans often:

- spend years trying to prove  $c$ ,
- fail,
- then begin believing  $\neg c$ .

Your procedure never does that. At every stage:

- proving  $c$  and
- proving  $\neg c$

receive equal computational attention.

Moreover, because all derivations arise from the same theory  $T$ , any lemma discovered on either side can be reused later. This is important. You are not really maintaining two separate proofs. You are maintaining a growing derivation graph generated from  $T$ . Some nodes happen to lie on paths leading toward  $c$ . Some nodes happen to lie on paths leading toward  $\neg c$ . The machine does not privilege either destination.

That is a philosophically attractive notion of “open-minded ATP search.”

## Relation to Clay-type Problems

For something like the conjecture  $P \stackrel{?}{=} NP$ , the machine would output one of:

- $T \vdash P = NP$ ,
- $T \vdash P \neq NP$ ,
- neither found within  $N$  steps.

The third outcome is unavoidable because Gödel-style incompleteness tells us that some statements may be independent of the chosen axiom system.

So a long-running search cannot distinguish:

- false,
- true,
- independent.

It only tells you that no proof has yet appeared.

## How This Fits Into Depths of a Simulation

This is where I think your framework becomes more distinctive. Let

- $S_c(N)$  = simulation attempting to settle  $c$  by the symmetric search procedure.
- $d(c)$  = minimal simulation depth required before either a proof of  $c$  or a proof of  $\neg c$  appears.

Then one could define: The settlement depth of a conjecture  $c$  is the least depth at which a simulation witness produces a proof of either  $c$  or  $\neg c$ . Formally:

$$d(c) = \min \{d : \exists \text{ simulation witness of depth } d \text{ settling } c\}.$$

The key idea is that the simulation is not trying to prove  $c$  specifically. It is trying to settle  $c$ . That distinction matters.

## Settlement Depth Principle

A possible principle in your framework could be:

*The intrinsic difficulty of a conjecture is measured not by the shortest proof of the conjecture, but by the smallest simulation depth required to settle it in either direction.*

In that view:

- easy theorem  $\Rightarrow$  small settlement depth,
- easy refutation  $\Rightarrow$  small settlement depth,
- independent statement  $\Rightarrow$  potentially unbounded settlement depth.

## A Stronger SIC-style Conjecture

One could even formulate something like:

For every statement  $c$ , the minimal settlement depth is equal to the minimal depth of a simulation witness whose output determines the truth value of  $c$ .

That would shift attention away from proofs and toward simulations that generate proofs. In your terminology, the object of study would no longer be merely

$$T \vdash c,$$

but rather

$$\text{SimulationWitness}(d) \Rightarrow (T \vdash c) \vee (T \vdash \neg c).$$

The depth  $d$  then becomes a complexity measure of the conjecture itself.

## One Possible Obstacle

A logician would immediately ask: “What if  $c$  is independent of  $T$ ?”

For example, relative to some theory  $T$ ,

$$T \not\vdash c \quad \text{and} \quad T \not\vdash \neg c.$$

Then your symmetric ATP never settles the statement. In the Depths framework, this suggests a natural extension:

- finite settlement depth if a proof or refutation exists,
- infinite settlement depth if the statement is independent of the theory.

That gives a clean trichotomy:

- proof found,
- refutation found,
- depth diverges.

This feels very compatible with the way your SIC framework already treats increasing simulation depth as a resource.

## Overall Assessment

My impression is that the strongest formulation is not the ATP procedure itself—that part is close to classical dovetailed proof search—but the idea of defining a conjecture’s settlement depth as the minimal simulation depth needed to obtain either a proof or a refutation. That notion seems much more specific to the “Depths of a Simulation” program and could potentially be formalized as a new invariant attached to mathematical statements.

## Relation to the Preorder on SICs

**Question:** How does this relate to there being a preorder on SICs?

**Answer:** It relates quite naturally. As I understand your framework, an SIC (Simulation-Isomorphism Class) is represented by simulation witnesses, and there is a preorder on SICs arising from the notion that one simulation can realize, emulate, or occur within another at greater depth.

Let me denote the preorder by

$$A \preceq B,$$

meaning roughly that SIC  $A$  can be simulated by SIC  $B$  (or that  $B$  is at least as deep/powerful as  $A$  in the sense of your framework).

Then your conjecture-settling procedure can be viewed as assigning a computational task to an SIC.

## Settlement as an SIC Property

Fix a theory  $T$  and conjecture  $c$ .

Define

$$\text{Settles}(X, c)$$

to mean: SIC  $X$  contains a simulation witness which eventually outputs either a proof of  $c$  or a proof of  $\neg c$ .

Now the preorder immediately implies a monotonicity principle: If  $X \preceq Y$  and  $X$  settles  $c$ , then  $Y$  settles  $c$  as well.

The reason is simple. If  $Y$  can realize every simulation available in  $X$ , then  $Y$  can realize the settlement simulation that already existed in  $X$ .

Thus settlement is upward-closed in the preorder.

## Principal Upper Sets

For a fixed conjecture  $c$ , consider

$$U_c = \{X : \text{Settles}(X, c)\}.$$

Because of monotonicity,

$$X \in U_c \quad \text{and} \quad X \preceq Y \quad \implies \quad Y \in U_c.$$

So  $U_c$  is an upper set (upward-closed subset) of the SIC preorder. This is already interesting. Different conjectures determine different upper sets in the SIC universe.

## Minimal Settlement SICs

Suppose there exists an SIC  $M_c$  such that  $\text{Settles}(M_c, c)$  and every SIC that settles  $c$  lies above it:  $M_c \preceq X$  for all  $X \in U_c$ .

Then  $M_c$  would be the minimal SIC capable of settling  $c$ .

This is very close to your settlement-depth idea. Instead of measuring settlement by a number, one measures it by position in the preorder.

## Order-theoretic Complexity

In fact, this may be richer than a single numerical depth. Two conjectures might have:

- identical minimal depths,
- but incomparable minimal SICs.

For example, one conjecture may require a certain kind of simulation structure, while another requires a different structure. Then neither minimal SIC simulates the other.

In preorder language:

$$M_c \not\preceq M_d \quad \text{and} \quad M_d \not\preceq M_c.$$

That would mean the conjectures have different kinds of simulation complexity rather than merely different amounts.

## Connection with Your Quotient Poset

You have previously asked about the quotient

$$(\text{SIC}/\sim, \preceq),$$

where  $\sim$  identifies mutually simulable SICs.

In that quotient, conjecture settlement corresponds to an upward-closed region of the poset. A conjecture  $c$  determines:

$$U_c = \{[X] : X \text{ settles } c\}.$$

Then one can ask:

1. Does  $U_c$  have a least element?
2. Does it have several minimal elements?
3. Is it generated by an antichain?
4. Are there conjectures whose settlement region is the entire poset?
5. Are there conjectures whose settlement region begins only above some very high level?

These become purely order-theoretic questions about the SIC poset.

## A Particularly Interesting Possibility

One could define a preorder on conjectures themselves:

$$c \leq_{\text{SIC}} d$$

iff every SIC that settles  $d$  also settles  $c$ . Equivalently,

$$U_d \subseteq U_c.$$

Interpretation: Settling  $d$  is at least as hard, from the SIC perspective, as settling  $c$ .

This turns the preorder on SICs into a derived preorder on mathematical statements. That strikes me as one of the more promising directions, because it ties together:

- the SIC preorder,
- simulation witnesses,
- settlement procedures,
- conjecture complexity,

all within the same order-theoretic framework. The settlement algorithm then becomes not merely a search procedure but a way of probing where a conjecture sits relative to the structure of the SIC preorder.

## Answers to Order-Theoretic Questions

**Question:** What are your answers, with full justifications, to the questions below? Also can you generate any new theorems or describe any additional insights based on your answers when coupled with my “Depths of a Simulation” manuscript?

Assuming only the framework we’ve discussed (a preorder of SICs, a conjecture  $c$ , and the upward-closed settlement region  $U_c$ ), some of those questions can actually be answered as theorems, while others depend on structural properties of the SIC preorder itself. I’ll state what I think can already be proved.

**Theorem 1** (Settlement Regions are Upper Sets). Let

$$U_c = \{X : \text{Settles}(X, c)\}.$$

If  $X \in U_c$  and  $X \preceq Y$ , then  $Y \in U_c$ .

*Proof.* If  $X$  contains a simulation witness that settles  $c$ , and  $Y$  can simulate  $X$ , then  $Y$  can simulate that witness. Hence  $Y$  also settles  $c$ . Therefore  $U_c$  is upward closed.

This theorem is essentially immediate from the definition.  $\square$

### Question 1: Must $U_c$ Have a Least Element?

**Answer:** No.

**Counterexample:** Consider a poset consisting of two incomparable SICs  $A, B$  together with everything above them. Suppose both  $A$  and  $B$  settle  $c$ . Then

$$U_c = \uparrow A \cup \uparrow B.$$

No least element exists because neither  $A \preceq B$  nor  $B \preceq A$ .

Therefore the existence of a least settlement SIC is not automatic.

### Question 2: Can $U_c$ Have Several Minimal Elements?

**Answer:** Yes. In fact this should be expected. The previous example already shows it. Both  $A$  and  $B$  are minimal members of  $U_c$ . Thus

$$\text{Min}(U_c) = \{A, B\}.$$

**Interpretation:** There may be several fundamentally different ways to settle the same conjecture.

### Question 3: Is Every Settlement Region Generated by an Antichain?

**Answer:** Yes, under a mild assumption. Let

$$M_c = \text{Min}(U_c).$$

Minimal elements are automatically pairwise incomparable. Therefore  $M_c$  is an antichain. Moreover,

$$U_c = \uparrow M_c$$

whenever every element of  $U_c$  lies above some minimal element. This holds in many well-founded posets.

So in a well-founded SIC universe, every settlement region is generated by an antichain. This is probably the most important structural observation. A conjecture may correspond not to a single minimal simulator but to an entire antichain of minimal simulators.

#### Question 4: Can the Settlement Region Be the Entire Poset?

**Answer:** Yes. Take a trivial proposition such as  $0 = 0$ . Every sufficiently expressive simulation can certify it. Thus

$$U_c = P$$

(the whole SIC poset).

More generally, if every SIC contains enough machinery to derive  $c$ , then  $U_c = P$ .

#### Question 5: Can Settlement Begin Only Above a High Level?

**Answer:** Yes. This is exactly where your framework becomes interesting. Suppose there is a chain

$$D_0 \prec D_1 \prec D_2 \prec \dots$$

representing increasing simulation depth. Assume  $c$  is not settleable below level 1000, but becomes settleable at level 1000. Then

$$U_c = \{D_n : n \geq 1000\}.$$

This is a perfectly legitimate upward-closed set. Thus conjectures can possess threshold depths.

**Theorem 2** (Settlement Threshold Theorem). Suppose the SIC quotient poset is totally ordered by depth. Then every settlement region has the form

$$[d, \infty)$$

for some depth  $d$  (possibly infinite).

*Proof.* In a chain, every upper set is an interval of that form. Therefore settlement of a conjecture corresponds exactly to crossing a depth threshold.  $\square$

This theorem strongly resembles phase transitions.

**Theorem 3** (Independence Theorem). Let  $c$  be independent of the underlying theory. Then no proof-search simulation settles  $c$ . Consequently

$$U_c = \emptyset.$$

*Proof.* Neither  $c$  nor  $\neg c$  is provable. Therefore the dovetailed ATP never halts. Hence no settlement witness exists.  $\square$

# Settlement Types and New Invariants

This gives four possible settlement types:

- **Empty region**  $U_c = \emptyset$  (independence)
- **Proper upper region**  $\emptyset \subsetneq U_c \subsetneq P$  (nontrivial theorem/refutation)
- **Principal upper set**  $U_c = \uparrow x$  (unique minimal settlement SIC)
- **Entire poset**  $U_c = P$  (trivial/easily decidable)

## New SIC Invariant

This is the insight I find most promising. Instead of assigning a numerical depth to a conjecture, assign its minimal-settlement antichain

$$M_c = \text{Min}(U_c).$$

Call this the *settlement spectrum* of  $c$ .

Then:

- Empty spectrum corresponds to independence.
- One-element spectrum corresponds to a unique minimal simulation type.
- Multi-element spectrum corresponds to several irreducibly different settlement mechanisms.

This is richer than a single depth. Two conjectures could have identical minimum depths but completely different settlement spectra.

## Connection to Your “Depths of a Simulation”

The strongest synthesis I see is this:

*Depth is only the one-dimensional shadow of a more fundamental order-theoretic object.*

The truly intrinsic object attached to a conjecture is not a number but its settlement region  $U_c$ .

Depth becomes a derived invariant:

- minimum depth,
- maximum depth,
- depth distribution,
- density of settlement SICs,

all extracted from  $U_c$ .

In that picture, your SIC preorder plays a role analogous to the role that the Turing degrees play in computability theory.

A conjecture is no longer classified merely by “how hard it is,” but by where in the simulation universe settlement first becomes possible and by how many incomparable routes to settlement exist. That seems like a genuinely new direction suggested by the interaction between the SIC preorder and your conjecture-settlement procedure.